

Differentiable Minkowski Functionals for Neural Fields: Validation, Design Rules, and an Adversarial Failure Mode

Gunner Levi Howe
Independent Researcher
gunnerlevihowe@gmail.com

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Abstract

The Minkowski functionals of a field’s excursion sets—area, boundary measure, and Euler characteristic—are the complete integral-geometric description of its level-set morphology, and the Euler characteristic is the cheapest handle on its *topology*. We derive smooth Monte-Carlo estimators for all three functionals of a continuous neural field, evaluated at scattered sample points via the co-area formula and the Gauss–Bonnet theorem, using only automatic differentiation: no grid, no simplicial or cubical complex, no persistence computation. The Euler-characteristic estimator is accurate to 1–3% against exact topology in 2D and 3D (bumps, merges, holes, handles) at 2×10^5 samples, and costs ~ 3 ms per training iteration where a persistent-homology (PH) loss on a cubical grid costs ~ 650 – 1000 ms—a $250\times$ gap. We establish four design rules without which these losses silently fail: a dense *level ladder* (topological invariants are almost-everywhere flat in the parameters; gradients exist only near transitions), a C^2 *backbone* (ReLU networks hide curvature in kinks invisible to sampling), the full *Minkowski vector* (the Euler characteristic alone is an alternating sum and is gamed by debris-hole cancellation; pricing perimeter closes the channel), and *sampling-scale coverage*. In 2D, the resulting vector-valued cap is the only method in a controlled comparison that both repairs topology (3/3 seeds) and preserves signal fidelity—uniform smoothing repairs at 11–17 $\times$ the fidelity cost, and the Euler term alone repairs nothing. In 3D neural-SDF fitting, however, we measure a failure mode we believe generalizes to any sampled soft topology objective: gradient descent adversarially concentrates topological noise *below the sampling density*, where the estimator and its guards are provably blind—spurious-feature counts are invariant to a $4\times$ increase in samples, and closing the blind window requires cubically many points, erasing the cost advantage. A grid-based PH baseline, whose complex *is* the evaluation resolution, solves the same benchmark (4/9 exact topology; median b_1 error 1 vs. ours $> 10^4$). We conclude that the $250\times$ cost of persistence is, at present, the price of having no null space, and we release estimators, receipts, and benchmarks.

1 Introduction

Controlling the topology of a learned implicit shape—“one component, two handles, no cavities”—is a recognized open problem: the persistent-homology (PH) literature itself notes that classical invariants such as Betti numbers “are not easily used” as direct controls in implicit surface reconstruction [3], and the state of the art constrains topology through persistence diagrams computed on cubical complexes at every training step [12]. That machinery is powerful but expensive and grid-bound.

Integral geometry offers a lighter-looking route. For a smooth field $f : \Omega \subset \mathbb{R}^d \rightarrow \mathbb{R}$ and excursion sets $A_u = \{f \geq u\}$, the Minkowski functionals—in 2D: area $M_0(u)$, boundary length $M_1(u)$, Euler characteristic $M_2(u) = \chi(A_u)$ —admit *integral representations* that require neither a grid nor a complex. This paper asks, and answers with pre-registered gates and kill conditions, the natural question: can those representations be turned into differentiable training objectives for neural fields, and do they displace persistence?

This is the third study in a program that turned the Kac–Rice level-crossing density (exactly M_1) into a spectral-bias loss for implicit neural representations [7] and a one-sided stabilizer budget for physics-informed networks [6]. The answer here has the same honest shape as those papers: the machinery is real, validated, and fast; its working domain is real but bounded; and the boundary itself is the most instructive result.

Table 1: Estimator validation (3 seeds, $N = 2 \times 10^5$, $\varepsilon = 0.02$). Fields decay below probed levels at the domain boundary, so no Gauss–Bonnet boundary terms arise.

Field	exact χ	$\hat{\chi}$ range (3 seeds)	s.d.
2D one bump	1	0.998–1.021	≤ 0.012
2D two bumps (merge)	2 / 1	2.005–2.017 / 1.006	≤ 0.026
2D annulus (hole)	0	0.011–0.019	≤ 0.040
3D ball	1	0.987–1.012	≤ 0.019
3D solid torus	0	0.016–0.026	≤ 0.071

Contributions. (1) *Estimators*: smooth Monte-Carlo Minkowski profiles of continuous neural fields at scattered points (Section 2), with the Euler characteristic via a Gauss–Bonnet/co-area integrand in 2D and 3D, validated to 1–3% against exact topology (Table 1) at ~ 3 ms per training iteration. (2) *Design rules*, each discovered through a documented failure: the level ladder, the C^2 backbone, the Minkowski-vector defense, and sampling-scale coverage (Section 3). (3) *A controlled 2D success*: the vector-valued cap is the only method in a four-way comparison that repairs topology on every seed while preserving fidelity (Table 2). (4) *A measured 3D failure mode*: sampled topology losses are adversarially exploitable in their sub-sampling null space; the debris is invariant to affordable increases in sampling, while a faithful PH baseline solves the benchmark at $250\times$ the per-iteration cost (Table 3, Figure 1).

What we do not claim. Differentiable Euler-characteristic machinery exists for discrete complexes [11], statistical shape transforms [8], and pixel grids [9]; PH losses for implicit surfaces exist [3, 12]; curvature energies for neural SDFs exist [5, 10]. Our claim is confined to the continuous-field, complex-free, scattered-point instantiation—and we show its current 3D limit as prominently as its 2D success.

2 Method

Estimators. With a Gaussian bump $\delta_\varepsilon(z) = (2\pi\varepsilon^2)^{-1/2}e^{-z^2/2\varepsilon^2}$ and samples $x_1, \dots, x_N$ covering Ω (volume V), the co-area formula gives

$$\hat{M}_0(u) = \frac{V}{N} \sum_i \sigma((f(x_i) - u)/\varepsilon), \quad \hat{M}_1(u) = \frac{V}{N} \sum_i \delta_\varepsilon(f(x_i) - u) \|\nabla f(x_i)\|, \quad (1)$$

and, for the Euler characteristic, Gauss–Bonnet on the level set: in 2D, $\chi(A_u) = \frac{1}{2\pi} \oint_{\partial A_u} \kappa_g ds$ with $\kappa_g = -\operatorname{div}(\nabla f / \|\nabla f\|)$, hence

$$\hat{\chi}(u) = -\frac{V}{2\pi N} \sum_i \delta_\varepsilon(f(x_i) - u) \frac{f_{xx}f_y^2 - 2f_{xy}f_xf_y + f_{yy}f_x^2}{\|\nabla f(x_i)\|^2}, \quad (2)$$

and in 3D, via $\int_S K dA = 2\pi\chi(S)$ and $\chi(\text{solid}) = \chi(S)/2$ with Goldman’s implicit-surface Gaussian curvature [4], $\hat{\chi}(u) = \frac{V}{4\pi N} \sum_i \delta_\varepsilon(f_i - u) \nabla f_i^\top \operatorname{adj}(H_i) \nabla f_i / \|\nabla f_i\|^3$. All derivatives come from automatic differentiation; the sign conventions are fixed by the unit-disc case and verified in released unit tests. Losses match or one-sidedly cap the profile vector at L fixed levels with the relative normalization of the companion papers.

Validation. Table 1: on analytic fields with hand-verifiable excursion topology—including the sign-critical annulus ($\chi = 0$) and solid torus ($\chi = 0$)—the estimator is exact to within 1–3% with seed-level standard deviations below 0.08, in both dimensions, at $N = 2 \times 10^5$. As a *measurement*, the tool is settled.

3 Four design rules, four documented failures

Each rule below was forced by an experiment that failed without it; the receipts (run logs and seeds) ship with the code.

Table 2: 2D topology repair (3 seeds; grid-verified). “Repaired” means (1 component, 0 holes) at every probed level.

Config	repaired	spurious max	clean MSE
anchor only	0/3	0.698 ± 0.092	0.0284 ± 0.0196
+ smoothness (null)	2/3	0.129 ± 0.010	0.0392 ± 0.0262
+ χ -cap alone	0/3	0.355 ± 0.219	0.0057 ± 0.0037
+ Minkowski-vector cap (ours)	3/3	0.109 ± 0.001	0.0041 ± 0.0010

Rule 1: the level ladder. $\chi(A_u)$ is a topological invariant, hence piecewise-constant in the network parameters; a smoothed estimator supplies gradients only within $\sim \varepsilon$ of a topology-changing event. A cap probing levels $[0.1, 0.4]$ against a spurious feature peaking at 0.7 received *exactly zero loss for 2,000 iterations*. Dense levels with spacing $\lesssim \varepsilon$ spanning the value range form a ratchet: the topmost active level always sits near the offending peak and supplies a live extinction gradient.

Rule 2: the C^2 backbone. On a ReLU network (with smooth positional encoding) the almost-everywhere Hessian omits the distributional curvature concentrated at activation kinks; the sampled Gauss–Bonnet integral then systematically under-counts and the loss never activates. On a SIREN backbone the same estimator reads the same field’s topology correctly. Smooth activations are a requirement, not a preference—consistent with standard practice for curvature losses [5, 10].

Rule 3: the Minkowski vector. χ is an alternating sum, so a χ -cap is satisfiable by *cancellation*: in our runs the optimizer manufactured four components and three holes ($\chi = 1$, as demanded) rather than deleting the one spurious component. Capping M_1 simultaneously prices the cancellation channel—debris costs perimeter—after which honest deletion is the cheapest descent path. The vector, not the invariant, is the loss.

Rule 4: sampling-scale coverage. All three estimators are blind to features smaller than the sample spacing. Rule 4 is the one that cannot be satisfied affordably in 3D, and Section 5 measures the consequence.

4 Controlled 2D comparison: the vector cap works

A SIREN field is polluted with a spurious component, then repaired under an anchor on clean-signal samples plus one of four regularizers; ground truth topology is read from a 256^2 grid, never from the estimator itself. Table 2: the anchor alone never repairs; Hessian smoothing repairs 2/3 but pays 11–17 $\times$ in fidelity when it does (same-seed MSE ratios 17.0 and 11.5; it smears the signal along with the blob); the χ -cap alone repairs 0/3 (Rule 3’s cancellation, observed on every seed); the vector cap repairs **3/3** with the best or near-best fidelity throughout, and drives the spurious region below the lowest probed level. In 2D—where 12k samples resolve everything a 256^2 evaluation can see—the method does exactly what it promises.

5 3D neural SDFs: an adversarial failure mode

Benchmark. SIREN SDFs fitted to 2,000 noisy surface samples with 5% uniform outliers, for a sphere, torus, and genus-2 solid; Eikonal and anchor losses throughout; topology read as voxel Betti numbers at 96^3 (verified exact on the analytic shapes); Chamfer distance to the clean surface; 3 seeds. Baselines: Eikonal only, Hessian smoothing, our vector match, and a faithful cubical-complex PH loss in the STITCH spirit [12] via GUDHI [13], differentiable through its critical-cell values. Two regimes: a hard one ($\omega_0=15$, 8k collocation points) and a bandwidth-matched one ($\omega_0=8$, Eikonal weight 1.0, 16k points). The PH configuration runs at half the iterations—its per-iteration cost is $\sim 250\times$ ours—and both matched-iteration and matched-wall-clock readings favor the conclusions below.

Table 3: 3D SDF benchmark (3 shapes $\times$ 3 seeds per regime; “topo correct” = exact Betti numbers (b_0, b_1, b_2) at 96^3). PH runs at half iterations; its topo cost is per-iteration.

Config	topo correct	median b_1 err	Chamfer	topo ms/iter
<i>hard ($\omega=15, 8k$) regime</i>				
Eikonal only	0/12	390	0.239	0.0
+ smoothness	0/12	276	0.221	0.2
+ Minkowski vector (ours)	0/12	5734	0.275	3.1
+ persistent homology	1/12	4	0.188	828.3
<i>matched ($\omega=8, 16k$) regime</i>				
Eikonal only	0/9	161	0.204	0.0
+ smoothness	0/9	81	0.210	0.1
+ Minkowski vector (ours)	0/9	91390	0.256	2.7
+ persistent homology	4/9	1	0.260	654.3

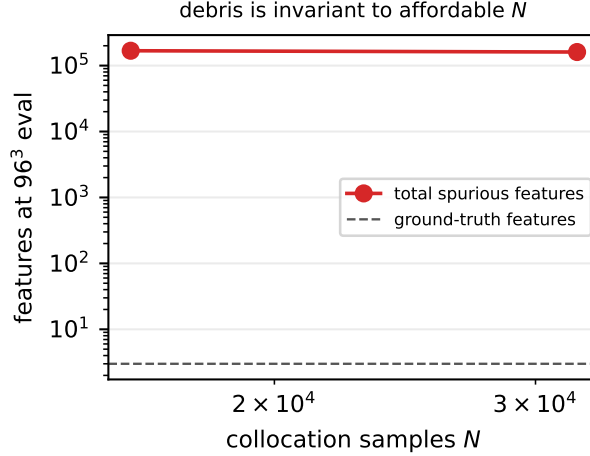


Figure 1: The blind window does not close affordably. Total spurious features of the vector-match SDF (median over shapes, seed 0, matched regime) versus collocation samples N : flat where a sampling-resolution mechanism would require decay, because the debris relocates below whatever spacing is affordable.

Result. In the hard regime every method fails topologically—including PH (1/12)—establishing that the benchmark is genuinely difficult. In the matched regime PH separates: 4/9 exact (3/3 on the sphere) with median b_1 error 1. Our vector match does not merely fail there; it fails *spectacularly*. Throughout, “spurious features” means the summed Betti excess $b_0+b_1+b_2$ over ground truth at the 96^3 evaluation: the vector match accumulates $\sim 1.7 \times 10^5$ in total, of which $\sim 7 \times 10^4$ are components (b_0) and the table’s median b_1 error alone exceeds 9×10^4 —one field, three views of the same debris.

Diagnosis: the null space is found, not stumbled into. Three facts localize the mechanism. First, the debris is loss-induced: Eikonal-only runs show 2–3 orders of magnitude fewer features. Second, the debris lives at the voxel scale (~ 0.02), strictly below the collocation spacing (0.05–0.08): precisely where $\hat{\chi}$, and the M_1 guard of Rule 3, receive no samples and hence no gradient. Gradient descent, asked to satisfy a topology target it cannot honestly reach quickly, discovers the estimator’s blind window and hides topological noise inside it—an adversarial attack by the optimizer on its own objective. Third, and decisive: the debris count is *invariant* to a $4\times$ increase in samples (Figure 1); spacing falls from 0.08 to 0.05 while the window to 0.02 remains open. Closing it requires $N \sim (2/0.02)^3 \approx 10^6$ points, whose Hessian evaluations would consume the entire $250\times$ cost advantage. The grid-based PH loss has no such window—its complex *is* the evaluation resolution—and that, we conclude, is what its cost currently buys.

6 Related work

Differentiable Euler-characteristic transforms operate on simplicial/cubical data [11]; smooth Euler characteristic transforms serve statistical shape inference [8]; differentiable Minkowski surrogates exist for pixel grids via relaxed morphology [9]. Topology control for implicit surfaces is dominated by persistence [3, 12]. Curvature-based (non-topological) energies for neural SDFs include the Eikonal term [5], its divergence-stabilized refinement [14], and zero-Gaussian-curvature CAD priors [10]. On the theory side, the *expected* Minkowski functionals of Gaussian-related fields are the subject of the Gaussian kinematic formula [1, 2]—the same random-field-theory lineage as the Rice formula that seeded this program; our estimators are its deterministic, per-field, trainable counterpart. The co-area/Gauss–Bonnet Monte-Carlo route for *continuous neural fields at scattered points*, and its use as a training loss, is to our knowledge new here—as is the negative half of its characterization. The estimators extend the Kac–Rice program of [6, 7], whose M_1 machinery this paper reuses verbatim as the Rule-3 guard.

7 Discussion and limitations

When to use which tool. As a *measurement* on smooth fields, the Minkowski profiles are accurate, differentiable, dimension-agnostic, and essentially free; we expect them to be useful as diagnostics and evaluation statistics well beyond losses (the companion papers’ crossing-profile diagnostic is the M_1 special case). As a *training loss*, the validated domain is: smooth backbone, dense level ladder, full vector, and sampling that covers the evaluation scale—conditions met in our 2D setting and unmet, affordably, in 3D. Where they cannot be met, persistence on a complex at evaluation resolution remains the tool of record.

Open routes for 3D. The failure is a sampling-allocation problem, not an estimator error: uniform collocation wastes almost all samples away from level sets. Importance sampling concentrated near $\{f \approx u\}$, multi-resolution ladders, debris-seeking adversarial samplers, or hybrid schemes (cheap Minkowski steps punctuated by sparse PH corrections) are all plausible; none are tested here, and we make no claims about them.

Limitations. Boundary terms are dodged by construction (fields decay at the domain edge); levels are fixed rather than adaptive; the PH baseline is our faithful-in-spirit reimplement rather than released STITCH code; shapes are analytic solids rather than scanned geometry; and all wall-clock figures come from one workstation.

8 Conclusion

We set out to displace persistent homology with integral geometry and did not. What stands instead: a validated, 250×-cheaper family of differentiable topological observables for neural fields; four transferable design rules, each carrying the failure that forced it; a controlled setting where the approach is the only one that works on both axes; and a precisely measured adversarial mechanism—optimizers hunt the null space of sampled objectives—that we believe any future sampled topology loss must answer. The boundary between the geometry that sampling can police and the topology that requires a complex is now measured, not conjectured.

Reproducibility. Code, unit tests (sign conventions, exact-topology validation, gradient flow), all raw run JSONs, and the failure-receipt logs: <https://github.com/gunnerhowe/Research> (directory `kac-rice`; estimators in `src/kacrice/minkowski.py`, SDF harness in `src/kacrice/sdf.py`, suites in `experiments/run_phase3.py`, figures via `experiments/make_figures_phase3.py`). Environment: Python 3.13.6, PyTorch 2.7.1, NumPy 2.4.4, SciPy 1.17.0, GUDHI 3.13.0, scikit-image 0.26.0; one consumer workstation (16-core CPU / RTX 3080). Seeds 0–2 throughout (seed 0 for the sampling sweep); every table regenerates from `results/phase3/*.json`.

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